

TEXAS LOADS RIDE TOWARD GRID STABILITY

Voltage Ride Through of Large Power Electronic Loads

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Introduction

IMAGINE SEEING AN EMPTY FIELD DE-void of any infrastructure. Now imagine returning one year later and, instead of an empty field, there are five incredibly large warehouses. Each is filled floor to ceiling with thousands of computers, all humming together in a loud cacophony, busy solving cryptocurrency or maybe artificial intelligence (AI) calculations. This structure maybe consumes a massive 300 MW; many larger ones are under construction. Told from the perspective of the Electric Reliability Council of Texas (ERCOT), this article explores the unique set of challenges large power

electronic loads (PELs) bring and the possible solutions to better integrate these large loads into the electricity grid.

ERCOT is one of several independent system operators (ISOs) in the United States ([Figure 1](#)). As the ISO for the region, ERCOT manages the flow of electric power to more than 27 million Texas customers, representing about 90% of the state's electric load. ERCOT's mission is to serve the public by ensuring a reliable grid, efficient electricity markets, open access, and retail choice. Thus, ERCOT needs to determine the best way to manage new challenges, whether past-to-date integration of more than 64 GW of renewables or the ongoing integration of incredibly large, rapidly developing PELs.

Digital Object Identifier 10.1109/MPE.2025.3564963

Date of current version: 21 August 2025

Currently, there are 57 GW of large loads, each more than 75 MW, requesting interconnection to ERCOT (Figure 2). Roughly three fourths are either data centers or otherwise computer based. This is an almost unimaginably large addition to the bulk electric system, which currently has a peak load of approximately 85 GW. While it is assumed that not all will be built, the rapid growth of these loads has spurred significant changes in the way that ERCOT plans and operates the electric system.

Traditionally, the impact of load growth is assessed in annual and/or ad hoc studies that strictly adhere to North American Electric Reliability Corporation (NERC) reliability standards and ERCOT transmission planning criteria. The rapid growth in large loads has led to a push for faster interconnection timelines. ERCOT established the Large Flexible Load Task Force to identify the measures needed to address the operational, planning, and market impacts of interconnected large loads in the ERCOT region.

In March 2022, through this task force, ERCOT implemented an interim process to ensure compliance with ERCOT and NERC requirements for large loads wishing to interconnect to the ERCOT system on an accelerated timeline.

ERCOT considers any large load as subject to the interim process. A 75-MW threshold is used to classify whether a load is “large.” This does not correspond to another requirement; rather it is the level at which ERCOT considers it prudent for a load to undergo a formal review and the threshold at which trans-

mission studies are more likely to identify a need for transmission upgrades.

What Are Large PELs?

PELs utilize switching semiconductors to process and convert electricity. Many modern loads utilize power electronics for efficiency or flexibility. Because many types of loads natively require dc to operate, the power electronics will often include a *rectifier* to convert ac power to regulated dc power. Typical large loads, such as computer data centers, hydrogen electrolyzers, and industrial motor drive

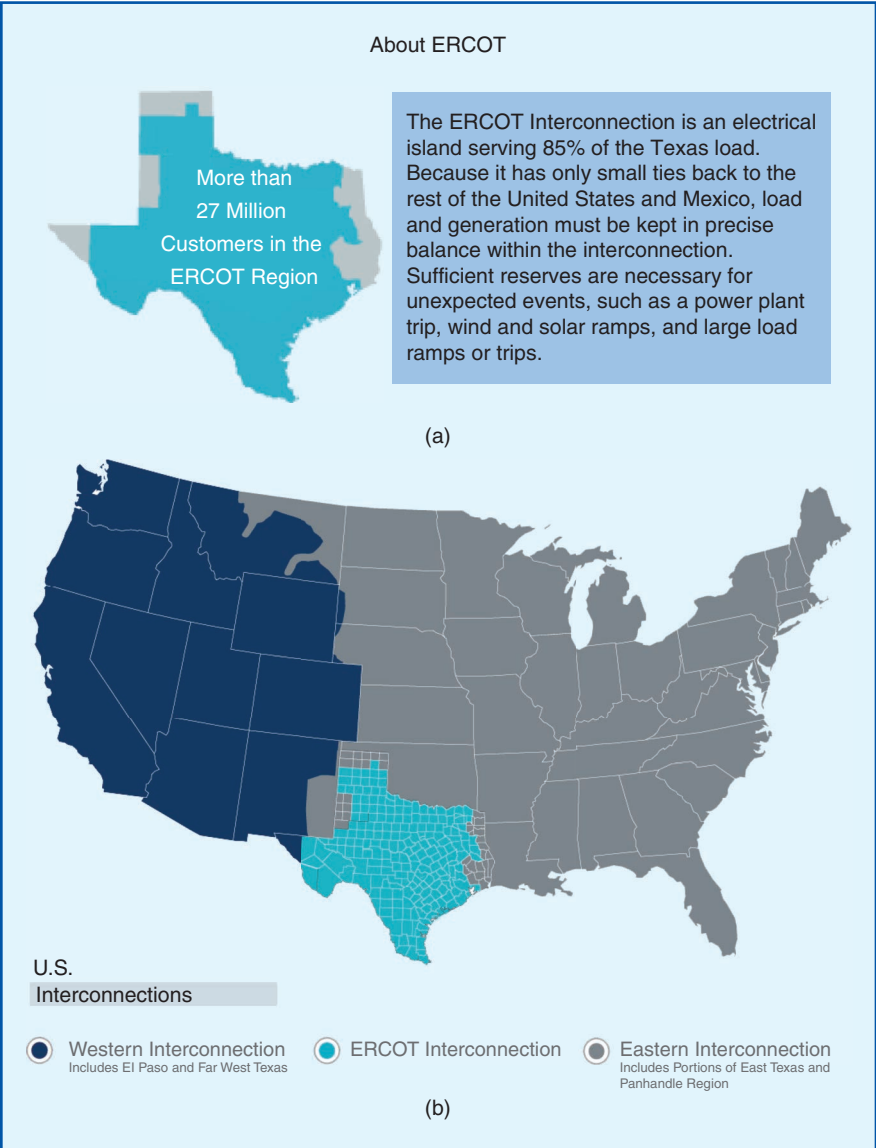


Figure 1. The ERCOT grid is one of three interconnections in the United States. The ERCOT corporation functions as an ISO to manage the ERCOT grid, serving as a balancing authority, planning authority, and reliability coordinator. ERCOT is one of several ISOs in the United States.

facilities, all utilize dc power. The first two utilize dc power directly, while the third usually incorporates dc as an intermediate stage before variable frequency inversion. In addition to rectification, a *power factor correction (PFC)* circuit is also normally needed to reduce harmonics and reactive power strains and increase efficiency.

There are generally two broad categories of rectifiers (Figure 3). Rectifiers have relatively simple controls, or, in the case of the inexpensive passive diode rectifier, they may have no controls at all. This is in contrast to the dc-to-ac inverters used in wind turbines, solar photovoltaic, and energy storage systems, which have complicated phase-synchronizing and voltage-feedback controls that can, on rare occasions,

misbehave in unexpected ways. The lack of complicated feedback controls on the grid side possibly means that rectifier-based loads can be represented by relatively simple, approximately algebraic models in most time-domain power system stability studies. The word “possibly” is used because this algebraic hypothesis has not been fully proven; additional laboratory testing and system event benchmarking is necessary. (To learn more about complicated power electronics behaving in unexpected ways on the power system, refer to the “Odessa Disturbance” listed in the section “For Further Reading.”)

Note that the power conversion equipment can be incorporated into a larger control system, for example, an industrial load governed by product orders

or a cryptocurrency data center load governed by the real-time market price of electricity relative to the monetary reward for mathematically validating blockchain transactions.

The dc bus ripple-smoothing capacitor in Figure 3 plays an important role in the voltage ride-through capability of the device, particularly in the case of computer power supplies. For economic reasons, this capacitor is typically quite small, only sized so that the converter can ride through a momentary drop in voltage (such as one cycle, or 1/60th of a second in

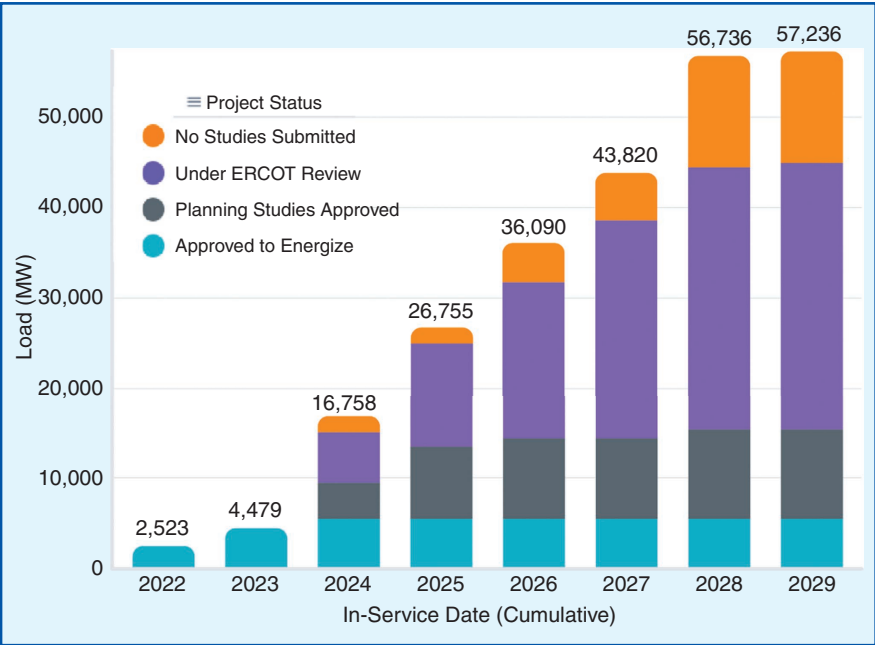


Figure 2. Actual and projected large load growth for 2022–2029 as of August 2024.

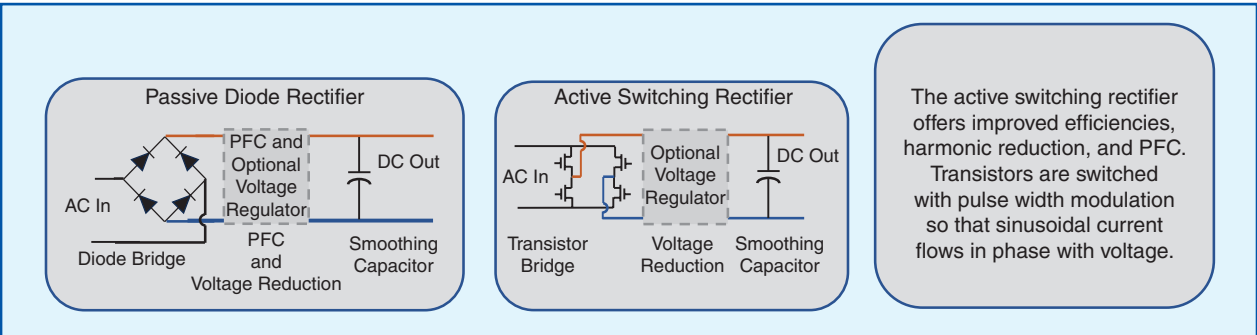


Figure 3. Two types of rectifiers commonly used in modern PELs. Both types efficiently convert ac to dc power. A detailed understanding is not necessary for grid studies but helps justify modeling assumptions.

North America) without interrupting the dc output and causing the computer to reboot. This is not well aligned with the four-to-seven cycles typically required for protective elements to clear a fault on the electric grid. As a result, nearby fully loaded data centers will likely reboot if they do not have backup *uninterruptible power supplies (UPSs)* on site. Many cryptocurrency data centers do not have such UPS devices, as their work is tolerant of interruptions, while many AI and cloud computing data centers do include UPS devices.

A typical computer power supply, such as the kind used in data centers, has additional components to step down and regulate the final output voltage (Figure 4) in addition to protection logic to shut down should the dc output voltage exceed strict tolerances. Many other dc loads, including hydrogen electrolyzer facilities, may have a subset of similar components.

The possibility that several massive data centers could reboot or switch over to UPSs at the same time for the same nearby grid fault, causing thousands of megawatts of load to momentarily disappear for a few minutes, has numerous reliability implications and concerns.

Power System Stability

Introduction to Power System Stability Studies

A variety of disturbances can impact the performance of a power system. Large voltage disturbances, called *faults*, involve electricity being temporarily shorted to ground and may be caused by

The frequency could be affected, export stability limits could be exceeded, and a sudden change in flows could cause overloads or voltage violations.

lightning, animal intrusion, or equipment failure, for example. Time-domain simulations are performed to simulate such disturbances and ensure that the power system returns to a stable operating point within acceptable tolerances.

The integration of PELs into the electric power system introduces several stability challenges that must be studied.

- ✓ *Voltage stability*: The substantial and variable demand from PELs can cause voltage instability. Effective voltage regulation and sufficient reactive power reserves are necessary to prevent disruptions.
- ✓ *Frequency stability*: Maintaining the grid frequency within acceptable limits is critical. The addition of large PELs can complicate frequency management, requiring advanced control strategies and sufficient reserves.
- ✓ *Loss of load events*: Operational processes, such as procedures for the procurement of additional reserves or postdisturbance operating plans, must be in place to cope with potential overvoltage and/or overfrequency when PELs are suddenly removed from the grid.

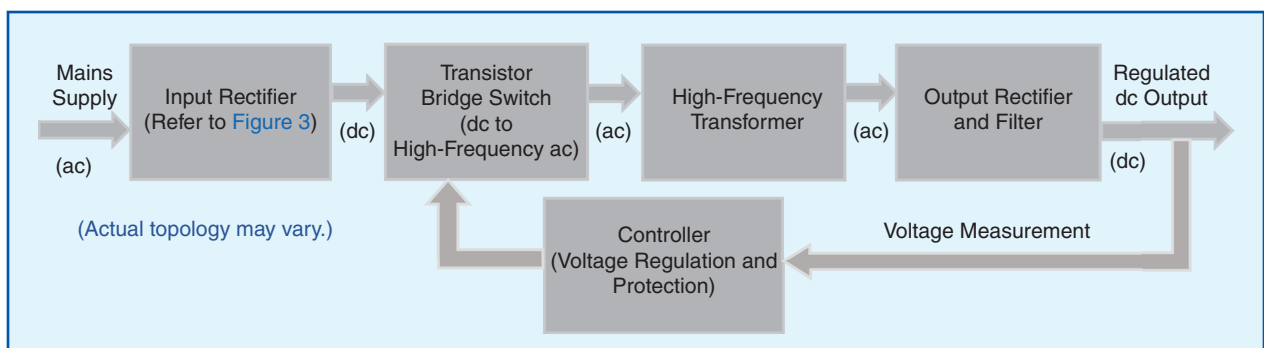


Figure 4. Topology of a typical ac-to-dc converter, such as those used in computer power supplies. The size of the capacitors within the rectifiers generally determines the voltage disturbance ride-through capability.

If many large loads in an area were to behave similarly, their tripping could cause voltage excursions and power imbalance between generation and load.

The accuracy of power system studies depends heavily on the collective accuracy of the input models. Unfortunately, the precise characteristics of large loads are often not well known when facilities are in the planning phase. Of special importance is whether the load under study, along with nearby loads, will trip for a given event. To address the tripping uncertainty, ERCOT requires running load trip sensitivities when they are perceived to have an impact on stability. For the load modeling issue, research groups have provided broad starting points for certain categories of load, which can be modified by the load customer based on available data to conservatively account for unknowns. While these measures provide a workable stopgap, a better understanding of site-specific load characteristics is very much needed.

The next several paragraphs discuss possible modeling assumptions for PELs. These assump-

tions are based on ERCOT's understanding of the load functioning rather than on information directly relayed by load customers. Direct information would be superior, but ERCOT has observed PEL customers having substantial difficulty obtaining useful information from their equipment suppliers—perhaps because the large PEL industry is still relatively new, and only recently were these devices installed in such large quantities that one needed to be concerned about possible grid impacts.

A PEL Model for Power System Studies

It is common to model electronic loads—particularly data centers—as constant-power loads down to the voltage trip point. Computer power supplies and similar voltage-regulated ac-to-dc converters contain fast regulation circuits that will compensate for grid disturbances, allowing the dc load to be unaffected by moderate grid events. This effectively decouples the computer from many grid dynamics and reinforces constant-power operation. In the event of more severe dips, the device will shut down to avoid damage. Computers generally adhere to the Information Technology Industry Council (ITIC) curve (Figure 5) for uninterrupted operation. Most faults on the electric grid are cleared when protection elements detect and remove faulted

equipment, typically in four to seven cycles (1 cycle = 1/60th of a second). This means that the ITIC curve specifies uninterrupted operation only if the supply voltage remains above 70%. The voltage could be much lower near the fault as power is temporarily sourced to ground. This could potentially lead to interruption at nearby PELs and even PELs more than 100 miles away.

Data centers are made up of thousands of such computers. For the most part, the single-load model can be scaled up.

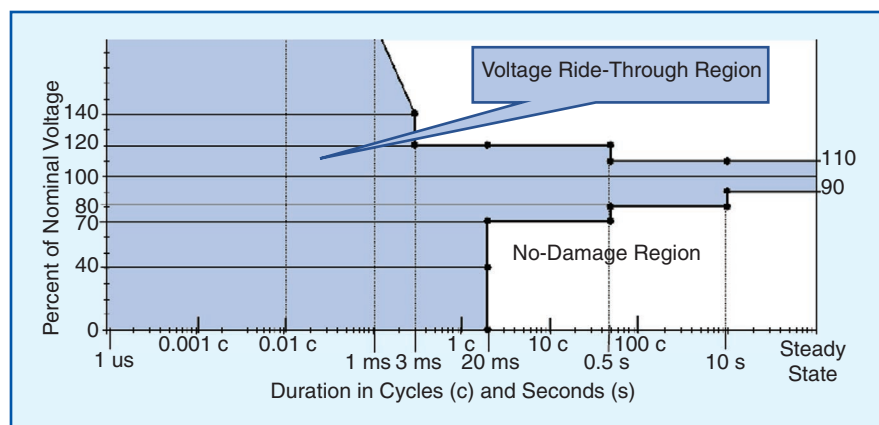


Figure 5. The ITIC curve for computers and other IT equipment. Although technically not a ride-through requirement, computer power supplies are generally designed to successfully ride through the low-voltage portion without interruption. A compliant computer running at full load will generally survive one cycle at 0 V without rebooting. For a typical transmission fault cleared in four to seven cycles, the voltage at the computer's electrical outlet must be above 70% to avoid a reboot.

However, the trip threshold will be higher because of the voltage drop in the facility power distribution wiring. The voltage drop tends to increase under low voltages because of the increased current flow necessary to serve a constant-power load. Thus, a 70% voltage trip threshold at the computers may correspond to a 75% or higher voltage at the point of grid interconnection.

Figure 6 is a suggested data center load model adapted from the Electric Power Research Institute electronic load model. The percentages of each type of load can be customized by input from the facility owner.

Further work is still needed to improve load models. The constant-power model does not consider electromagnetic effects like harmonic and subsynchronous interaction. Power losses and the effect of voltage drop within the facility are not well quantified. Some facilities will consume additional power once the voltage recovers as motors speed back up and dc capacitors and batteries recharge. Some non-computer PEL loads may not exhibit constant-power behavior. UPS systems may introduce additional characteristics, altering the dynamic performance.

The Voltage Ride-Through Problem

Self-disconnection of a single load may not have much impact on the system; however, the simultaneous disconnection of hundreds or even thousands of megawatts of large loads in an area could pose a risk to reliability. The frequency could be affected, export stability limits could be exceeded, and a sudden change in flows could cause overloads or voltage violations.

An Actual Loss of Load Event

At 3:50 a.m. on 7 December 2022, multiple related faults occurred on 138-kV lines in

a rural part of the ERCOT system, including one three-phase fault due to a circuit breaker failure. The event induced a 1,600-MW load reduction. Loads that tripped during this event included data centers, oil/gas pumping facilities, and other industrial facilities that were unable to ride through the extended low-voltage period. Additionally, some generation was lost, including two thermal generators totaling 112 MW. As a result of the loss of load, the system frequency spiked to 60.235 Hz and returned to 60 Hz in 12 min, 30 s. Figure 7 shows the system frequency spike response, and Figure 8 shows the voltage dropping to almost half of its nominal value at a nearby 345-kV station for this event.

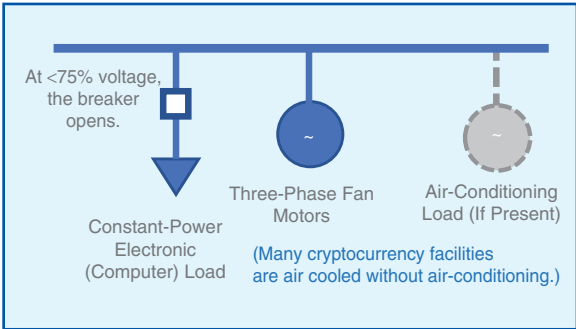


Figure 6. Example equivalent stability model for a typical data center in grid stability studies.

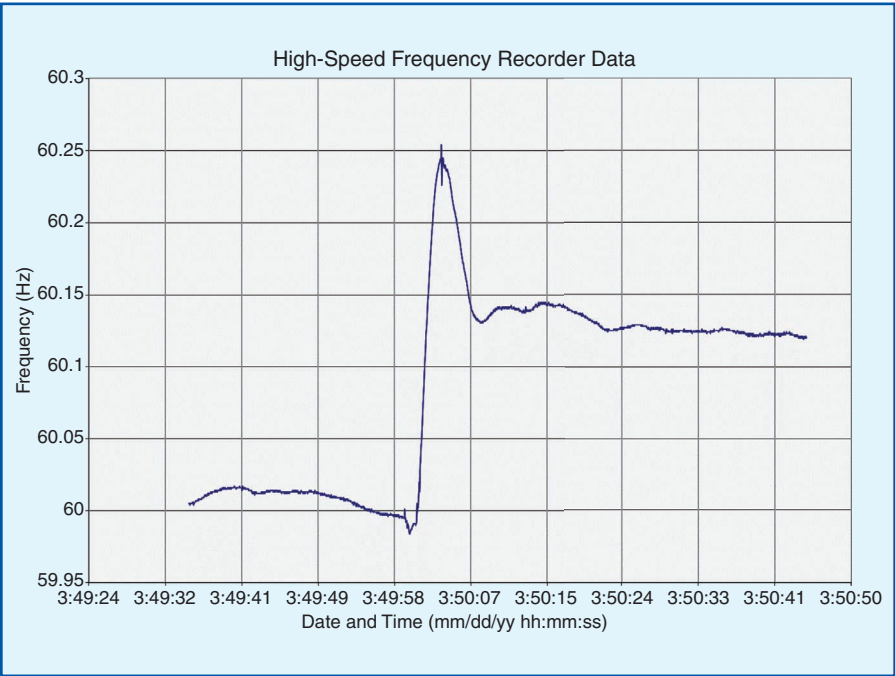


Figure 7. System frequency: loss of load event on 7 December 2022.

While the event did not result in any substantially negative impacts, the sensitivity of large loads to voltage disturbances raised concerns. If many large loads in an area were to behave similarly, their tripping could cause voltage excursions and power imbalance between generation and load, and lead to frequency excursions and other operational problems.

Case Study: Impact of Large PELs on Grid Stability

Increased renewable development has caused stability constraints to appear in ERCOT's rich renewable resource areas. Because electrical instability can evolve very quickly, operators must take action ahead of time, meaning that system operators will constrain operating conditions in preparation for the next unplanned contingency. To manage this situation and other similar ones on the ERCOT grid, ERCOT uses generic transmission constraints (GTCs) to manage stability, voltage, and other constraints that cannot be directly modeled in ERCOT's online steady-state power flow and contingency analysis applications. A GTC is a tool that ERCOT operators use to geographically constrain the transmission flow on one or more defined interfaces (selected transmission elements determined by a study) to avoid unstable operating conditions. Stability is maintained by limiting the GTC interface flow below

a known specific limit, called the *generic transmission limit*. Integrating PELs behind a GTC can significantly impact the generic transmission limit.

The addition of large loads without an adequate voltage ride-through capability presents significant challenges in managing this constraint. For example, large loads have expressed interest in locating behind such stability constraints where electricity prices may be lower. Adding a load here permits the dispatch of additional generation because some generation will be absorbed locally, and the stability limit will be preserved. This works until a transmission fault causes a line to trip and the large load trips as well due to the brief voltage drop. In this case, the stability limit could be exceeded, sending the area into oscillations or other instability.

This challenging situation is illustrated in [Figure 9](#). The simplest management approach currently being implemented will reduce the stability limit by an amount needed to preserve stability whenever a load without assured ride-through capability is added behind a generation export GTC. This is not the most efficient outcome as adding a 500-MW load without ride-through capability could result in reducing up to 500 MW of generation on the system under certain scenarios.

Improving the Ride Through of PELs

ERCOT implemented a low-voltage ride-through standard for renewable generation in 2009. Implementing a similar standard for load remains challenging because most load equipment and facilities were not designed considering their impacts on the grid. Just as the renewable ride-through standard required some innovation to design wind turbines and solar inverters, a similar innovation will be required for large loads.

While efforts to implement ride-through standards for load has been met with hesitancy from industry, ERCOT has

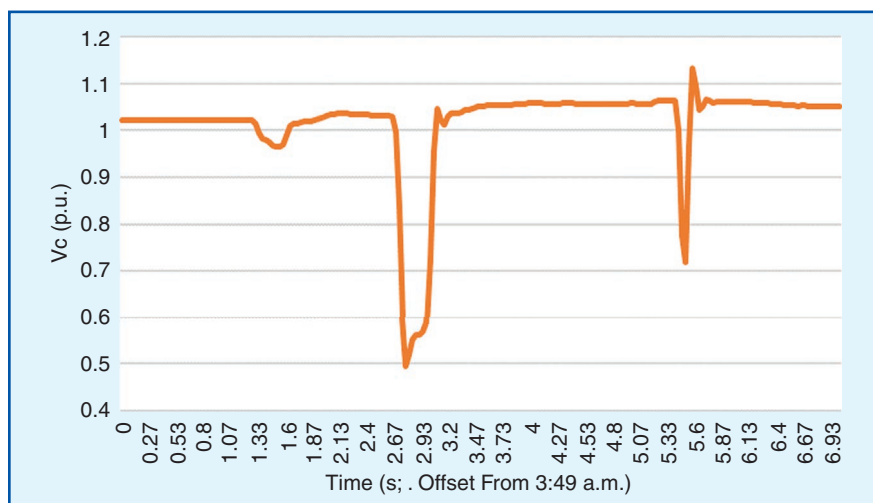


Figure 8. Voltage at nearby 345-kV station: fault and loss of load event on 7 December 2022.

been making progress engaging research and industry partners. Additionally, ERCOT has proposed a standard that limits the maximum amount of load that can be lost for a single credible planning event to maintain frequency stability (refer to the ERCOT Reliability Performance Criteria for Loss of Load in the section “For Further Reading”).

One significant challenge to overcome when considering how to improve the ride through of large loads—particularly large PELs—is the downstream process that the PEL is driving. Consider a motor driving an industrial process or a computer running important software code. If the power is interrupted, a lengthy restart process may be needed. It may be several minutes before the load ramps back to the predisturbance value. Thus, the best assurance seems to involve a method that avoids process interruption to begin with. But how can this be accomplished for a load of several hundred megawatts in size? Figure 10 illustrates some options the authors have gathered from research institutions; however, we are not experts in this field, and further collaboration is needed.

Critical loads often employ a UPS—essentially, a battery backup inverter system that picks up the load when the power fails. While this keeps the process running, it does not necessarily keep the load on the grid as many UPS systems are programmed to delay switching back until a period of observation determines acceptable grid integrity. While the period of observation is typically tens of seconds, the good news is that UPS systems are very programmable, perhaps providing a unique opportunity for the facility owner to coordinate with the utility operator.

Idea 2 of Figure 10 is appealing in that the technology exists today and is in widespread use.

Idea 3 is especially appealing because it makes PELs more grid friendly—by

Operators must take action ahead of time, meaning that system operators will constrain operating conditions in preparation for the next unplanned contingency.

modulating the load power proportional to voltage, the load changes from a constant-power load to more like a constant impedance. As the grid voltage drops, power reduces, and voltage recovery becomes much easier. A new IEEE standard, “Recommended Practice for System Stability of AC Power Converters in Data Centers” (IEEE Standard P3380), appears to be working on a similar goal.

Impact of Load Ramp Rates

PELs can exhibit very high ramp rates, from 0% to 100% consumption power in less than a second. Too high a ramping rate could create problems with overloads, voltages, and frequency stability. This may especially be a concern when a large number of PELs ramp according to a common signal, such as the price of electricity. It may become necessary to implement ramp rate restrictions for

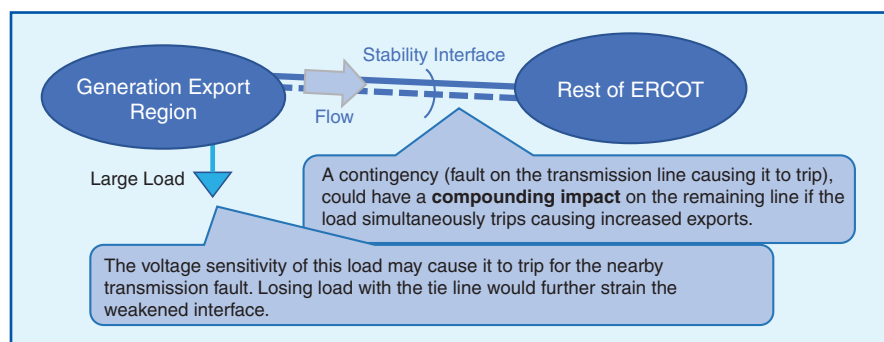


Figure 9. Example of a counterintuitive situation in which loss of a new large load can negatively impact and worsen an existing stability constraint. Here, two transmission lines connect a generation export region to demand in the rest of ERCOT. To avoid instability, the flow through the interface must be constrained below a predetermined level to avoid instability. Because generation redispatch takes time to implement and instability can happen very quickly, flow must be constrained precontingency. In this example, adding a large load inside the export region could worsen the stability constraint if it does not ride through. The stability interface limit may need to be lowered as a result of the load addition because the load may trip and send an excess of generation to flow down the already-strained transmission path.

very large loads. Until then, the worst-case impact of load ramping can be studied by either tripping or by reconnecting large loads in planning stability simulations.

Impact on Traditional Grid Operation and Planning

Incorporating large PELs into the traditional grid introduces several challenges and necessitates adjustments in planning and operational strategies. The following ERCOT concerns are based on past experiences or those potentially occurring in the near term.

Unplanned fluctuations in large PELs can complicate the ability of transmission operators to meet underfrequency load-shedding obligations. For example, a single PEL may represent half of the load served by a small transmission operator. In such a case, the underfrequency load-shedding obligation for the transmission operator may double or be cut in half as the demand from the PEL fluctuates. Additionally, the unpredictable nature of PEL demand, particularly when aggregated, can result in sudden drops or spikes in load, making it difficult to maintain grid stability without rapid response mechanisms in place.

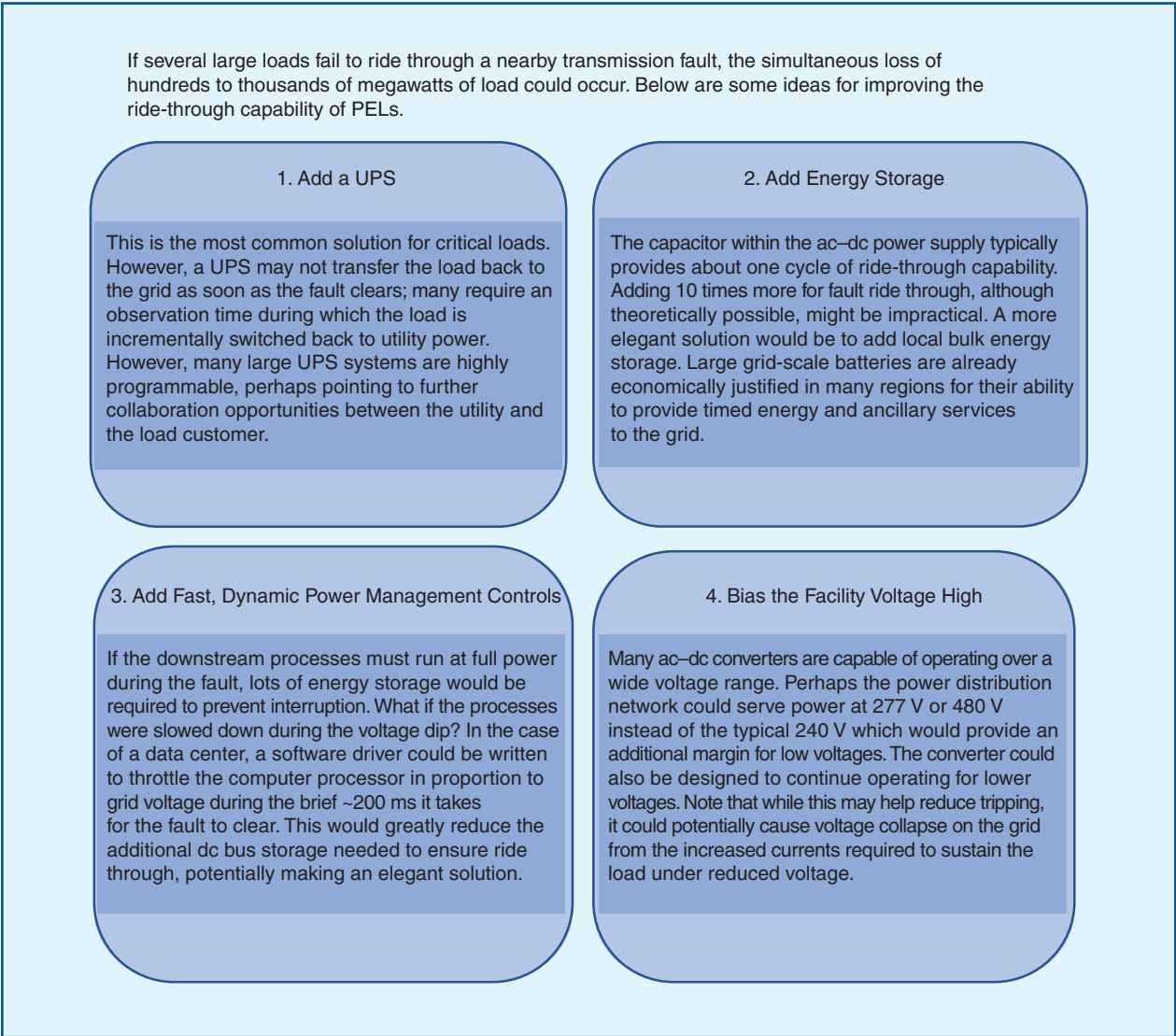


Figure 10. Potential solutions for improving the ability of PELs to ride through low voltages and transmission faults (further collaboration with industry is needed). Concepts originate from ERCOT discussions with large load industry partners, the Electric Power Research Institute, and other research institutions, such as Texas A&M University.

Large PELs can constitute a significant portion of the load in certain areas. This concentration of demand can stress the local transmission infrastructure and require substantial upgrades to accommodate the load without compromising reliability.

PELs often exhibit demand profiles that differ from those of traditional loads. For instance, they may operate continuously at high power levels or have significant loading flexibility depending on operational cycles and market conditions (e.g., cryptocurrency prices or electric market prices). This variability challenges traditional load forecasting and requires more dynamic and responsive grid management strategies.

ERCOT has identified several key milestones for incorporating large PELs into planning cases, including the following:

- ✓ *Initial interconnection request*: This is used to evaluate the feasibility and impact of the proposed load.
- ✓ *Detailed impact studies*: Thorough analyses are conducted to assess the effects on local and regional grid stability.
- ✓ *Integration planning*: Strategies can be developed to manage and mitigate identified impacts before the load is energized.

A summary of ERCOT's current guideline for large load studies is presented in [Figure 11](#).

Large Load Assumptions for Planning Studies

The 75% voltage trip threshold is being used as a placeholder until more accurate estimates or measurements can be obtained from PEL owners. Getting large load customers to provide validated models suitable for transmission studies is a significant challenge that requires greater industry collaboration. Historically, load customers have not needed to provide such models, and transmission companies would

estimate load behavior based upon surveys and statistical analyses of mixed and diffuse loads. These numbers would then get plugged into standard load models adjusted for the specific types of industry and the regional climate. While this routine works for diffuse loads where any unique behaviors average out, it does not appear adequate for large loads, which individually and collectively can have a significant stability impact.

Despite many being advertised as flexible and price responsive, most large loads currently do not register as a controllable load resource in the ERCOT market; thus, they are treated as nondispatchable in real time and likewise “firm” in planning studies. This “firm” designation makes it possible to use reliability as a justification to build transmission upgrades, but it also means that ERCOT may need to limit the size of the load interconnection if upgrades cannot be built in time. The Large Flexible Load Task Force has been considering potential modifications to the market rules with the intent to encourage more large loads to register as a controllable load resource dispatchable resource. This could provide a faster path for interconnection, especially if the constraint is expected to occur less frequently.

Once a study is completed, it may determine a need for upgrades. For example, in the case of voltage violations, it may be necessary to add dynamic

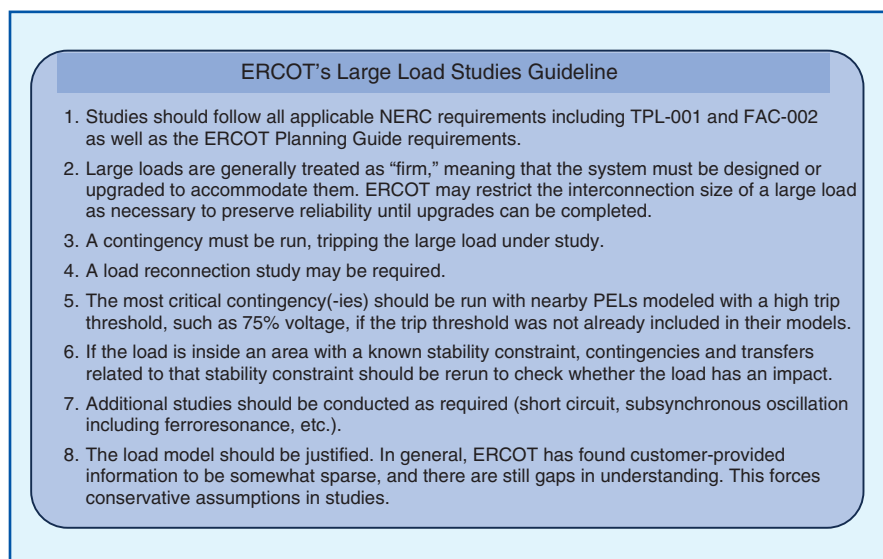


Figure 11. ERCOT's Large Load Studies Guideline for steady-state and dynamic stability studies.

ERCOT collaborates extensively with various stakeholders and research institutions to enhance grid reliability and address the challenges posed by PELs.

reactive transmission support for voltage control, such as adding static var compensators and STATCOMs.

Grid Modernization Initiatives

The old paradigm of loads being served by large, central generating stations through transmission networks and distribution systems is not sustainable in the future. PELs, such as data centers and cryptomining sites, can be developed much faster than the infrastructure that is needed to serve them. Whereas electric systems have typically been developed to simply serve loads in the past, loads in the future will generally need to be more integrated into system planning and operation practices. For example, mechanisms for leveraging load flexibility may be able to replace a need for additional infrastructure investment.

In the future, loads may incorporate battery-storage devices (or electric vehicle charging) that may offer opportunities to inject power into the grid. As a result, it will be important for load owners to have a better understanding of their load characteristics,

especially during grid disturbances, that they can share with grid planners and operators to ensure accurate and proper modeling for system studies. PELs may be adaptable to utilize advanced controls that provide a more grid-friendly performance characteristic. ERCOT has started several “grid modernization and enhancement” initiatives for continued research, development, collaboration, and innovation to address such evolving challenges and ensure grid reliability and resilience (Figure 12).

Collaboration With Stakeholders and Research Institutions

ERCOT collaborates extensively with various stakeholders and research institutions to enhance grid reliability and address the challenges posed by PELs:

- ✓ *Stakeholder engagement:* ERCOT engages with market participants, utility companies, and regulatory bodies to gather input and align on strategies for grid reliability. This includes regular meetings of working groups, public consultations, and collaborative focused planning sessions.
- ✓ *Research partnerships:* ERCOT has partnered with academic institutions to develop better models of data center load and to research ways to improve ride through.
- ✓ *Pilot projects:* ERCOT conducts pilot projects to test new technologies and methodologies in

real-world scenarios. These projects provide valuable insights and data that inform broader implementation strategies across the grid. One successful pilot project let energy storage resources participate in a fast frequency response program. This raised the level of comfort in dealing with large power unbalance disturbances, for example, when a large load trips.

Conclusion

The integration of large loads into the ERCOT grid brings

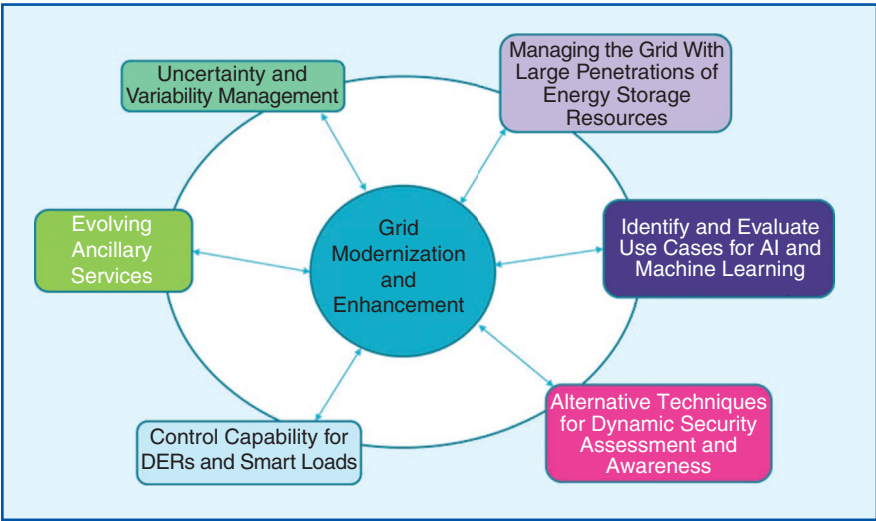


Figure 12. ERCOT grid modernization and enhancement initiatives. DER: distributed energy resource.

amplified risks to reliability, resiliency, and operating margins. The testing of the grid performance under multiple scenarios and contingencies becomes a necessity when the number of large PELs keeps increasing. This article highlights pressing problems that ERCOT faces when planning to interconnect multiple large loads, particularly PELs, in a relatively short period. We list some interim processes ERCOT has implemented to handle new large load interconnections, some of the challenges, like voltage ride through and effects on existing stability constraints, and also potential solutions, such as working with industry to obtain better models and improve voltage ride through and dynamic voltage characteristics.

ERCOT is committed to plan and maintain a reliable and stable grid, to provide an interconnection process of large loads that is fair, and to apply best practices and sound engineering when assessing the impact of large loads on the Texas grid. ERCOT is working with industry and research institutions to better understand the characteristics and hence the challenges of these very large additions to the system and is seeking collaboration to continue successful interconnections while ensuring reliability. Better modeling, improved load ride through, and better visibility in studies and operations are our continued goals.

Acknowledgment

The authors would like to thank Manuel Navarro Catalan, Chris Cosway, Julie Snitman, and Agee Springer with the ERCOT Grid Interconnections for providing review and feedback, and they also thank Patrick Gravois with the Operations Analysis team at ERCOT for gathering information about the load loss event.

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